Solving a Two–CRRA–Agent, Five–Tree Lucas Economy via Deep FBSDE with Malliavin–Gradient Regularization, Antithetic Coupling, and Adaptive RQMC

Technical Note

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Abstract

We present a practical recipe to compute equilibrium prices in a two–CRRA–agent endowment economy with five Lucas trees using a deep forward–backward stochastic differential equation (FBSDE) method. The approach leverages (i) Malliavin/differential regularization to stabilize the BSDE gradient process Z_t, (ii) antithetic path coupling, and (iii) adaptive randomized quasi–Monte Carlo (scrambled Sobol') sampling for variance reduction. We formulate the social planner, derive the state–price density (SPD), risk premia, and risk–neutral dynamics, then solve five linear pricing BSDEs for the tree prices under $\mathbb{Q}$. We provide training objectives that avoid free Y_0 tuning, implementation advice, and built–in diagnostics. A pentanomial (five–branch) tree discretization variant is noted.

1 Economic environment

Endowments (five Lucas trees). There are $K = 5$ dividend streams $D_t^k > 0$, each following a correlated Itô diffusion in $\mathbb{R}^5$ under the physical measure $\mathbb{P}$:

$$\mathrm{d}D_t^k \;=\; D_t^k\Big(\mu_k\,\mathrm{d}t + \boldsymbol{\sigma}_k^\top \mathrm{d}W_t\Big), \qquad k = 1, \ldots, 5, \tag{1}$$

where $W_t \in \mathbb{R}^5$ is a standard Brownian motion with instantaneous covariance matrix $\Gamma = I_5$ (correlations are embedded in the $\boldsymbol{\sigma}_k \in \mathbb{R}^5$ loading vectors). Stack dividends as $\boldsymbol{D}_t = (D_t^1, \ldots, D_t^5)^\top$. Aggregate consumption is the single good

$$C_t \;=\; \sum_{k=1}^{5} D_t^k. \tag{2}$$

Then

$$\mathrm{d}C_t \;=\; \mu_C(\boldsymbol{D}_t)\,\mathrm{d}t + \boldsymbol{\sigma}_C(\boldsymbol{D}_t)^\top \mathrm{d}W_t, \qquad \mu_C(\boldsymbol{D}) = \sum_{k=1}^{5} D^k \mu_k, \quad \boldsymbol{\sigma}_C(\boldsymbol{D}) = \sum_{k=1}^{5} D^k \boldsymbol{\sigma}_k. \tag{3}$$

Agents. Two infinitely–lived agents $i \in \{1, 2\}$ have time–additive CRRA utilities with common subjective discount rate $\rho > 0$:

$$U_i = \mathbb{E}\Big[\int_0^T e^{-\rho t}\, u_i(c_{i,t})\,\mathrm{d}t\Big], \qquad u_i(c) = \frac{c^{1-\gamma_i}}{1-\gamma_i}, \quad \gamma_i > 0,\ \gamma_i \neq 1. \tag{4}$$

Markets are complete (five risky assets span five shocks; no frictions). A competitive equilibrium is Pareto efficient.

Planner's problem. Fix Pareto weights $\lambda_1, \lambda_2 > 0$. At each t, the planner solves

$$\max_{c_{1,t}, c_{2,t}} \lambda_1 u_1(c_{1,t}) + \lambda_2 u_2(c_{2,t}) \quad \text{s.t.} \quad c_{1,t} + c_{2,t} = C_t. \tag{5}$$

FOC yield a common multiplier $\psi_t > 0$ such that

$$\lambda_i u_i'(c_{i,t}) = \lambda_i c_{i,t}^{-\gamma_i} = \psi_t, \qquad i = 1, 2. \tag{6}$$

Therefore $c_{i,t} = \big(\lambda_i/\psi_t\big)^{1/\gamma_i}$ and, using feasibility, $\psi_t = \psi(C_t)$ is defined implicitly by

$$C_t \;=\; \sum_{i=1}^{2} \Big(\frac{\lambda_i}{\psi_t}\Big)^{1/\gamma_i}. \tag{7}$$

When $\gamma_1 = \gamma_2 = \gamma$, the allocation is linear: $c_{i,t} = \alpha_i C_t$ with $\alpha_i = \lambda_i^{1/\gamma}/(\lambda_1^{1/\gamma} + \lambda_2^{1/\gamma})$, and

$$\psi_t \;=\; \Big(\lambda_1^{1/\gamma} + \lambda_2^{1/\gamma}\Big)^{\gamma} C_t^{-\gamma}. \tag{8}$$

State price density (SPD). Let M_t denote the SPD. In a time–additive planner with constraint (5), the Lagrange multiplier delivers

$$M_t = e^{-\rho t}\,\psi_t \;=\; e^{-\rho t}\,\psi(C_t). \tag{9}$$

Write the diffusion of C_t as above, and consider $\psi(C)$ as a C^2 function. Using Itô, define

$$\psi'(C) := \frac{\mathrm{d}\psi}{\mathrm{d}C}, \qquad \psi''(C) := \frac{\mathrm{d}^2\psi}{\mathrm{d}C^2}, \tag{10}$$

$$\mu_\psi(C, \boldsymbol{D}) := \psi'(C)\,\mu_C(\boldsymbol{D}) + \tfrac{1}{2}\psi''(C)\,\|\boldsymbol{\sigma}_C(\boldsymbol{D})\|^2, \qquad \boldsymbol{\sigma}_\psi(C, \boldsymbol{D}) := \psi'(C)\,\boldsymbol{\sigma}_C(\boldsymbol{D}). \tag{11}$$

Then the $\mathbb{P}$–dynamics of M_t is

$$\frac{\mathrm{d}M_t}{M_t} \;=\; -r_t\,\mathrm{d}t \;-\; \boldsymbol{\theta}_t^\top \mathrm{d}W_t, \quad \boldsymbol{\theta}_t \;=\; -\frac{\boldsymbol{\sigma}_\psi}{\psi}(C_t, \boldsymbol{D}_t) \;=\; -\frac{\psi'(C_t)}{\psi(C_t)}\,\boldsymbol{\sigma}_C(\boldsymbol{D}_t), \quad r_t \;=\; \rho - \frac{\mu_\psi}{\psi}(C_t, \boldsymbol{D}_t). \tag{12}$$

In the equal–risk–aversion case (8), these simplify to

$$\boldsymbol{\theta}_t \;=\; \frac{\gamma}{C_t}\,\boldsymbol{\sigma}_C(\boldsymbol{D}_t), \qquad r_t \;=\; \rho + \frac{\gamma\,\mu_C(\boldsymbol{D}_t)}{C_t} - \frac{\gamma(\gamma+1)}{2}\,\frac{\|\boldsymbol{\sigma}_C(\boldsymbol{D}_t)\|^2}{C_t^2}. \tag{13}$$

2 Risk–neutralization and asset–pricing BSDEs

Define the risk–neutral measure $\mathbb{Q}$ by the density M_t/M_0. Under $\mathbb{Q}$,

$$\mathrm{d}W_t^{\mathbb{Q}} = \mathrm{d}W_t + \boldsymbol{\theta}_t\,\mathrm{d}t, \tag{14}$$

and each dividend process satisfies

$$\mathrm{d}D_t^k \;=\; D_t^k\Big(\mu_k^{Q}(t, \boldsymbol{D}_t)\,\mathrm{d}t + \boldsymbol{\sigma}_k^\top \mathrm{d}W_t^{\mathbb{Q}}\Big), \qquad \mu_k^{Q}(t, \boldsymbol{D}) := \mu_k - \boldsymbol{\sigma}_k^\top \boldsymbol{\theta}_t. \tag{15}$$

Let P_t^k denote the ex–dividend price of tree k (terminal value $P_T^k = 0$, running dividend D_t^k). Its $\mathbb{Q}$–pricing PDE is linear:

$$\partial_t u^k + \mathcal{L}^{\mathbb{Q}} u^k - r\, u^k + D^k = 0, \qquad u^k(T,\cdot) = 0, \tag{16}$$

where $\mathcal{L}^{\mathbb{Q}}$ is the generator of $(\boldsymbol{D}_t)$ under $\mathbb{Q}$. The equivalent BSDE is, for $t \le T$,

$$Y_t^k \;=\; \int_t^T \Big(r_s\, Y_s^k - D_s^k\Big)\,\mathrm{d}s \;-\; \int_t^T \big(\boldsymbol{Z}_s^k\big)^\top \mathrm{d}W_s^{\mathbb{Q}}, \qquad P_t^k = Y_t^k. \tag{17}$$

We solve (17) for all $k = 1, \dots, 5$ simultaneously with a single network (multi–task heads).

3 Time discretization

Let $0 = t_0 < t_1 < \cdots < t_N = T$ with $t_{n+1} - t_n = \Delta t$. Denote path states by $\boldsymbol{X}_n := (\boldsymbol{D}_{t_n})$ and, when needed, augment with $C_n = \sum_k D_{t_n}^k$. Simulate $(\boldsymbol{X}_n)$ under $\mathbb{Q}$ using Euler–Maruyama:

$$D_{n+1}^k = D_n^k\Big(1 + \mu_k^Q(t_n, \boldsymbol{X}_n)\,\Delta t + \boldsymbol{\sigma}_k^\top \Delta W_n^{\mathbb{Q}}\Big), \qquad \Delta W_n^{\mathbb{Q}} \sim \mathcal{N}(\mathbf{0},\, \Delta t\, I_5) \;\text{ i.i.d.} \tag{18}$$

(Using $\mathbb{Q}$ ensures a linear BSDE drift for Y.)

Discretize (17) in residual form:

$$\mathcal{R}_n^k := \widehat{Y}_{n+1}^k - \widehat{Y}_n^k + \Big(r_n\, \widehat{Y}_n^k - D_n^k\Big)\Delta t - \big(\widehat{\boldsymbol{Z}}_n^k\big)^\top \Delta W_n^{\mathbb{Q}}, \quad n = 0, \dots, N-1, \tag{19}$$

with terminal condition $\widehat{Y}_N^k = 0$. Here $\widehat{Y}_n^k$ and $\widehat{\boldsymbol{Z}}_n^k$ are network outputs at $(t_n, \boldsymbol{X}_n)$.

4 Networks and *differential* (Malliavin) regularization

We parameterize a shared spatio–temporal representation $\Phi_\vartheta(t, \boldsymbol{X})$ with two task–heads per asset:

$$\widehat{Y}_n^k = h_{\phi_k}^{(Y)}\big(\Phi_\vartheta(t_n, \boldsymbol{X}_n)\big), \qquad \widehat{\boldsymbol{Z}}_n^k = h_{\psi_k}^{(Z)}\big(\Phi_\vartheta(t_n, \boldsymbol{X}_n)\big) \in \mathbb{R}^5.$$

Let $u_\Theta^k(t, \boldsymbol{X}) := h_{\phi_k}^{(Y)}(\Phi_\vartheta(t, \boldsymbol{X}))$ with $\Theta = (\vartheta, \{\phi_k, \psi_k\}_{k=1}^5)$. By the Malliavin/Markov relation for semilinear BSDEs, the true Z satisfies $\boldsymbol{Z}^k = \nabla_{\boldsymbol{X}} u^k \cdot \boldsymbol{\sigma}^{\mathbb{Q}}(\boldsymbol{X})$ where $\boldsymbol{\sigma}^{\mathbb{Q}}$ stacks the columns that map $\Delta W^{\mathbb{Q}}$ to $\mathrm{d}\boldsymbol{X}$. We introduce a *gradient consistency* penalty:

$$\mathcal{L}_{\text{Mall}} \;=\; \sum_{k=1}^{5} \sum_{n=0}^{N-1} \lambda_n \Big\| \widehat{\boldsymbol{Z}}_n^k - \Big(\nabla_{\boldsymbol{X}} u_\Theta^k(t_n, \boldsymbol{X}_n)\Big)\, \boldsymbol{\sigma}^{\mathbb{Q}}(t_n, \boldsymbol{X}_n) \Big\|^2, \tag{20}$$

with an annealed schedule λ_n (start large, decay).

5 Losses without free Y_0

We avoid a free Y_0 scalar by predicting Y at all times and penalizing both the terminal boundary and the pathwise BSDE residuals:

$$\mathcal{L}_{\text{BSDE}} \;=\; \frac{1}{B} \sum_{b=1}^{B} \sum_{k=1}^{5} \left[\underbrace{\big\|\widehat{Y}_N^{k,(b)} - 0\big\|^2}_{\text{terminal}} \;+\; \sum_{n=0}^{N-1} w_n \big\|\mathcal{R}_n^{k,(b)}\big\|^2 \right], \tag{21}$$

and the total objective is

$$\mathcal{L}_{\text{total}} \;=\; \mathcal{L}_{\text{BSDE}} + \mathcal{L}_{\text{Mall}}. \tag{22}$$

Here B is the mini–batch size, w_n are residual weights (e.g., uniform or linearly increasing towards maturity).

6 Variance reduction: RQMC and antithetics

Adaptive scrambled Sobol'. Replace i.i.d. Gaussians by a scrambled Sobol' net in $[0,1]^{5N}$, mapped via inverse normal to obtain $\Delta W_n^{\mathbb{Q}}$. Increase either the number of points or the scramble depth across epochs (*adaptive RQMC*). Use a fixed seed per epoch to stabilize the optimizer.

Antithetic coupling. For each Sobol' point U, add its antithetic $1-U$; equivalently, for Gaussian shocks pair $\Delta W_n^{\mathbb{Q}}$ with $-\Delta W_n^{\mathbb{Q}}$ pathwise. Average residuals within each pair before backpropagation. This greatly reduces variance in $\widehat{Y}$ and $\widehat{\boldsymbol{Z}}$.

7 Algorithmic recipe

Algorithm 1: Deep FBSDE solver for five tree prices with differential regularization, antithetics, and adaptive RQMC.

Input: Horizon T, steps N, batch size B, model params $\{\mu_k, \boldsymbol{\sigma}_k\}_{k=1}^5$, CRRA $\{\gamma_i, \lambda_i\}_{i=1}^2$, discount ρ.

Output: Trained Θ that maps $(t, \boldsymbol{X}) \mapsto \{P^k = \widehat{Y}^k, \widehat{\boldsymbol{Z}}^k\}_{k=1}^5$.

Precompute $\psi(C)$, $\psi'(C)$, $\psi''(C)$ either in closed form (if $\gamma_1 = \gamma_2$) or via a robust 1D root solver with automatic differentiation.

for *epoch* $= 1, \ldots, E$ **do**

- Generate B pairs of RQMC paths under $\mathbb{Q}$: for each pair, sample Sobol' point(s) in $[0,1]^{5N}$, scramble, map to $\Delta W_n^{\mathbb{Q}}$, and create antithetic $-\Delta W_n^{\mathbb{Q}}$.
- **for** *each path* $b = 1, \ldots, 2B$ **do**
  - Simulate forward $\boldsymbol{X}_0^{(b)} \to \boldsymbol{X}_N^{(b)}$ with (18), using r_n from (12) and μ_k^Q from (15).
  - Evaluate network u_Θ^k and $h^{(Z)}$ along the path to obtain $\widehat{Y}_n^{k,(b)}$, $\widehat{\boldsymbol{Z}}_n^{k,(b)}$.
  - Form residuals $\mathcal{R}_n^{k,(b)}$ via (19); compute the Malliavin penalty (20) with $\nabla_{\boldsymbol{X}} u_\Theta^k$ from auto–diff.
- **end**
- For each antithetic pair, average losses before gradient accumulation.
- Update Θ by minimizing $\mathcal{L}_{\text{total}}$ (22) (Adam or similar; gradient clipping recommended). Optionally anneal λ_n and increase RQMC resolution.

end

8 Recovering equilibrium objects and checks

After training, the price of tree k at state $(t, \boldsymbol{X})$ is $P^k(t, \boldsymbol{X}) = u_\Theta^k(t, \boldsymbol{X})$. The dividend yield and risk premia follow from (15), (12). Useful diagnostics:

1. **BSDE closure:** pathwise residual norms $\left\|\mathcal{R}_n^k\right\|$ are small and stable across time.

2. **Gradient consistency:** the Malliavin gap $\left\| \widehat{\boldsymbol{Z}}_n^k - (\nabla_{\boldsymbol{X}} u_\Theta^k)\, \boldsymbol{\sigma}^{\mathbb{Q}} \right\|$ decays during training.

3. **No–arbitrage:** Euler checks under $\mathbb{P}$: confirm $\mathbb{E}_t^{\mathbb{P}}[M_{t+\Delta t}(P_{t+\Delta t}^k + D_{t+\Delta t}^k)]/M_t \approx P_t^k$.

4. **Aggregation:** verify $\sum_{k=1}^5 P_t^k$ against the price of the aggregate consumption claim priced by a single BSDE with running payoff C_t.

9 Practical notes

- **State choice.** Use $\boldsymbol{X} = (\log \boldsymbol{D}, \log C)$ or $(\boldsymbol{D}, C)$. Including C explicitly stabilizes $\psi(C)$ and r_t evaluations.
- **Equal–γ fast path.** If $\gamma_1 = \gamma_2 = \gamma$, use the closed forms (8)–(13). This considerably simplifies r_t and $\boldsymbol{\theta}_t$.
- **Two–head vs. single–head.** A single head u_Θ^k with $\widehat{\boldsymbol{Z}}^k = (\nabla_{\boldsymbol{X}} u_\Theta^k)\, \boldsymbol{\sigma}^{\mathbb{Q}}$ *enforces* gradient consistency but may be memory–heavier; the two–head + penalty in (20) is often more stable early on.
- **Loss weights.** Start with $w_n \equiv 1$ and $\lambda_n = \lambda_0$ large (e.g., 10^{-1}), halve λ_n every few epochs once residuals plateau.
- **RQMC schedule.** Begin with a modest number of Sobol' points per epoch, then double every S epochs. Keep one scramble per epoch fixed to decouple optimizer noise from sampler noise.
- **Antithetic batching.** Average losses and residuals across each antithetic pair before backpropagation; this is more effective than simply doubling the batch size.
- **Gradient clipping & LR.** Clip global norm (e.g., at 1–5); use a warmup and cosine decay or step decay; monitor the Malliavin gap to tune λ_n.

10 Appendix A: Computing $\psi(C)$, $\psi'(C)$, $\psi''(C)$ for general $\gamma_1 \neq \gamma_2$

Let $a_i := \lambda_i^{1/\gamma_i}$. From (7), define $\phi(\psi) := \sum_{i=1}^2 a_i \psi^{-1/\gamma_i}$ so that $C = \phi(\psi)$. Then

$$\phi'(\psi) = \sum_{i=1}^2 \Big(-\frac{a_i}{\gamma_i} \Big) \psi^{-1/\gamma_i - 1}, \qquad \phi''(\psi) = \sum_{i=1}^2 \frac{a_i}{\gamma_i} \Big(\frac{1}{\gamma_i} + 1 \Big) \psi^{-1/\gamma_i - 2}. \tag{23}$$

By inversion,

$$\psi'(C) = \frac{\mathrm{d}\psi}{\mathrm{d}C} = \frac{1}{\phi'(\psi)} \;=\; -\left[\sum_{i=1}^2 \frac{a_i}{\gamma_i} \psi^{-1/\gamma_i - 1} \right]^{-1} < 0, \tag{24}$$

and

$$\psi''(C) = -\frac{\phi''(\psi)}{\big(\phi'(\psi)\big)^3}. \tag{25}$$

Plug into (12) to obtain r_t and $\boldsymbol{\theta}_t$. For equal γ, (8) yields $\psi'(C) = -\gamma\, \psi / C$ and $\psi''(C) = \gamma(\gamma+1)\psi/C^2$, recovering (13).

11 Appendix B: A pentanomial (five–branch) variant

If you intended "five–tree" as a per–step pentanomial discretization of $\mathcal{N}(0, \Delta t)$, replace RQMC by the symmetric scheme with increments $\{-2, -1, 0, 1, 2\}\sqrt{\Delta t}$ and probabilities $p_{\pm 2} = 1/6$, $p_{\pm 1} = 1/6$, $p_0 = 1/3$. This matches mean 0, variance Δt, and fourth moment $3\Delta t^2$. Antithetics are obtained by flipping each token $j \mapsto -j$ along the path. The BSDE loss (22) is unchanged.

12 Minimal checklist (TL;DR)

1. Specify $\{\mu_k, \boldsymbol{\sigma}_k\}_{k=1}^5$, $\{\gamma_i, \lambda_i\}_{i=1}^2$, ρ, T, N.
2. Precompute $\psi(C), \psi'(C), \psi''(C)$ (closed form if $\gamma_1 = \gamma_2$).
3. Build r_t and $\boldsymbol{\theta}_t$ from (12); get $\mu_k^{\mathbb{Q}}$ via (15).
4. Train a multi–head BSDE network with residual loss (21) + Malliavin penalty (20).
5. Use scrambled Sobol' + antithetics; anneal λ_n; monitor residuals and Malliavin gaps.
6. Read off $P^k = u_\Theta^k$ and validate with Euler/SPD checks.